FICTION

Astronomy for Young Australians

James Bonwick

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ASTRONOMY

FOR

YOUNG AUSTRALIANS.

BY JAMES BONWICK, F.R.G.S., AUTHOR OF “GEOGRAPHY OF AUSTRALIA,” ETC., ETC.

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PREFACE

TO

SECOND EDITION.

A cheap edition of the “Astronomy for Young Australians” having been desired, to meet the requirements of the Public Schools of the colonies, Mr. Bonwick respectfully submits the present issue to the favourable attention of Teachers.

ST. KILDA, November 1, 1866.

ASTRONOMY

FOR

YOUNG AUSTRALIANS.

A Fine ship was buffeting the waves, like a strong swimmer in his sport. The clouds dashed wildly to and fro, but left many wide spaces of blue to be dotted with stars. The sea was in high spirits, throwing up spray, as if to quench the bright lights that looked down so kindly upon it.

The vessel quite enjoyed the fun, and her prow bobbed in the current, and gathered up the foam to send it flying up the rigging, or leave it hissing and fuming by her sides. How skittish she was this evening, as the light clouds ran over her head, and the wind puffed her laughingly along! She was young and strong. Her timbers were tight; and her yards well braced. She had just left port, painted and clean, with a new suit of sails, and her copper as bright as a fresh-coined penny.

And where was she going? Laden with the treasures of English cotton and woollen mills, of iron-works, and other industries, she was away to the far-off land of Australia, on the other side of the round globe.

And whom had she got on board? There were sailors to manage the merry ship; there were passengers, going from the white cliffs of old England, to dwell in the gum

forests of the kangaroo home.

As the shore of Britain melted away in the distance, men and women hung over the bulwarks, dreaming of sweet vales they left behind, and sighing deeply as they thought of loved ones there. How long would it be before they saw those vales again, or smiled with friends beside that hearth of love!

The love of one’s country, as the land of our friends, the joy of freedom, the defence of the right and true, is a duty as well as a delight. And those born in the new land of Australia should cherish so beautiful and healthful a home, and help to make it happy in the virtues of its people.

One little intelligent fellow, about ten years old, hung closely against his mother’s dress, and caught hold of his father’s hand. James knew his parents must feel sad at leaving dear ones behind; and he inwardly resolved to be a great comfort to them now by being a good, loving son.

The last speck of land was gone, and every body turned round to the ship, preparing to make that a home. Mr. and Mrs. Marple thought of their only child, for others had been left to sleep in a church-yard far away. He was their hope and joy. They determined to add to his happiness, and secure him from the evil of idle ship-life, by improving his time and his mind.

This very evening, therefore, the kind father took James beside him in a quiet corner on deck. They spoke at first of aunts and uncles, cousins and friends. Then a plunge of a porpoise turned their eyes to the sea, or a scream of sea-fowl set them talking of natural history. A sober chat followed, and a moral lesson came.

A sudden unrolling of clouds brought out such a dazzle of starry splendour, that both gazed with delight upon the ever-wondrous heavens. It was then that Mr. Marple determined, among other subjects, to make the boy understand astronomy. This he was to do by leading the lad himself to observe; and, under his direction, to find out the laws of the universe himself.

A few stars were pointed out. There was the North Polar star, that keeps such a constant place. There were the gentle Pleiades. There was the beautiful Belt of Orion to the south. There was, too, though very near the waves to the southward, the bright Dog-star, _Sirius_.

DAILY MOTION OF THE EARTH.

The following dialogue took place one evening on deck:--

“Well, my little fellow, what is there which so attracts your attention? for you have scarcely moved for this last half-hour.

I have been looking at the stars, father.

What is so wonderful in them this evening?

There is something wonderful in them which I never noticed before.

What is that?

I will tell you. When I first came upon deck after tea, I saw a bright star rising like as if it came out of the water. While looking at its pretty light, it seemed to get higher and higher up the sky.

Did it run away from the other stars, and get a head of them?

No. But I thought it did at first, until I had watched, and then I found that the stars around about kept at the same distance from my bright star, and appeared to keep company with it while climbing the sky.

And what else did you see?

Afterwards I thought I would watch the beautiful Sirius. This seemed to be moving on, too. Then I looked at others. But they all appeared to be rolling along after one another.

What do you make of all this?

I can make nothing of it.

What puzzles you, my boy?

Several things. I cannot understand where all the stars are rolling to, why they all keep together so cleverly in their motion, or why they are moving at all.

Did you see any stars setting as well as rising?

Yes, father.

Well, as I want you to find out this subject yourself, I shall allow you to stay up later this evening, to give you time to make a few more observations. Now, follow your bright star a little further. Look at that blue one overhead, and trace his journey. Have your eyes upon Sirius, and the band of Orion.

I will, father, and thank you."

* * * * *

The boy was left at his star-gazing, and intently was his mind fixed upon his work. It was no vacant stare he gave at the heavens. He had an object before him.

The conversation was afterwards renewed by the father:--

"I should think you were sleepy, James.

No, father: the stars kept my eyes open.

How, now, does your star get on?

Look up there. He has got as far as that.

Where is the blue-coloured one?

O, that has gone down to the western edge.

What of my old friend Sirius, and his neighbour Orion?

They have been travelling the same road.

Well, you must stop star-gazing to-night.

But will you not explain the reason of this curious motion, my dear father, before I go down?

No, my lad, you must have another look at the stars to-morrow evening first."

The little fellow retired slowly and thoughtfully to his berth; and dreamed of stars and ships in mingled confusion. How he longed for the sun to leave off his shining! He never watched that orb so as he did that day! He saw it rise, ascend, descend, and set. When the short twilight was over, the little twinkling bodies came out one by one, as a few western clouds changed from gorgeous red to colours of a darker hue. He first distinguished Sirius, and then the band of Orion. The blue star took some time to come forth; but when it did, there it was right overhead, as it had been early the last evening. The bright star was again on the tip of the distant waves. He stared again. He remembered how he had left them all the night before, and now the stars seemed in their old places again. His kind father came up to him.

"O, father," cried the boy, "all my stars have gone back into their places again this evening.

Did you see them travel back as you saw them travel forward?

No. But there they are, though I do not know how they got there.

I think I saw you looking pretty often at the sun, to-day. Did you notice anything peculiar about his movements?

Yes. I noticed that he seemed to go the same road as my stars did the evening before.

But you saw him ascend in the east, rise nearly overhead, and then set in the west. You did not observe your stars do that.

No, father; but I have it. It took a long while for the sun to go all that distance; and I fancy that if I had watched my bright star as long, I should have seen that set in the west also.

Certainly you would.

Well, but how did they get back into their places again?

How will the sun manage to get on the east side, to-morrow morning?

Ah! I see. What will explain the movement of the stars will explain the movement of the sun.

I think you will find it so.

There is only one way by which I can account for this--they must all turn round the earth. Is that it, father?

Everybody used to think so.

But what surprises me is this; as the stars are turning round us, none get before or behind the others. They keep the same distance apart.

Do you not know that all the stars are stuck in a huge mass of blue stuff, called the sky, and that when the sky turns all of them must turn with it, and, unless they tumble out, they must keep in their places?

O, father, you are joking.

Well, then, if they are not joined together, why do they move so uniformly?

That I cannot tell.

No two of the stars are of equal distance from us. They are all scattered through space, like a lot of marbles in a scramble.

Then if they had to move round the world, I am sure they never would keep in such order, especially as some would have to go so much faster than others; for they have a much greater distance to go. They cannot turn round the earth in that way. But yet they seem to do so.

Yes. But just look forward. Do you see how the head of the vessel is dipping into the Bay of Biscay?

I do, father.

But do you notice that as the bow of the ship sinks and rises, all the stars to the southward seem to dance up and down?

Now I understand. Like as these stars appear to dance about according to the motion of the ship, so the apparent motion of the stars round the earth in twenty-four hours must be owing to the motion of the earth. The earth must turn over once a day. It is easier to imagine this, than to imagine such extraordinary motions to take place among the stars."

ROTUNDITY OF THE EARTH.

It was fine fun for the healthy, hearty English boy to be tossed about in the Great Atlantic. What a noble, towering wave was that which raised up the ship, but only to let it down in the deep trough of the sea, while its crest hung over it, as if to drive it to the bottom!

As the day passed off, and the sombre clouds thronged the western horizon, James got toned down, and became thoughtful and still. Then was his face turned from the waves to the stars, being never tired of watching their motions and admiring their beauty.

He had long been regarding the heavens one evening, when he turned, as he heard his father's step, and exclaimed--

"The more I study the stars, father, the more their motions bother me.

How is that, my son?

When we first left England I saw the Polar star and Great Bear a good height up in the heavens, toward the north, and Sirius low down toward the south.

Have they been playing any tricks with you?

I do not know that. But as we have been sailing towards the equator, I have observed the Great Bear going gradually down, and Sirius as gradually coming up. I am afraid I shall lose the Polar star altogether soon. And now I have another trouble. I do not know what to make of a number of new stars coming up from the southward.

Would it be owing to the shape of the earth, think you?

Aye, I never thought of that.

If we were sailing on a flat surface all this long way, would you see these ups and downs of the stars?

No, I think not. The truth is, it seems to me that we have been going up a hill, if the stars do not move.

That will account for your Great Bear going down; but what should cause these new stars to appear in the south?

Why, that looks as though we were getting up to the top of the hill, and should have to go down on the other side.

What shape, then, do you suppose the world ought to be, to account for these curious apparent motions of the stars?

I fancy it ought to be round, and that is what I have always been told it is, father.

I want to ask you whether you have noticed anything peculiar about the apparent daily motion of the stars. Do they all rise in the east and pass over to the west?

This is just the subject I wanted to speak about. I thought at first that they did so, and ought to do so. But I have since found out that this is not the case. They do not all describe the same sized circle.

What have you observed?

Some make a regular sweep across--others only rise to pop down again. Yet there is a more wonderful thing. The stars near the North Polar star perform their circle without going under at all. Then the greatest puzzle is, that the Polar star always keeps its place without any turning.

You do not think that the stars describe circles each day of different sizes according to their distances from the Polar star?

No. I feel sure that it is all a cheat of the eye, though I don’t see where the cheat is.

I thought you were quite settled in your mind that the apparent motion of the stars was owing to the revolution of the earth.

True, father; yet I see no connection between the movement of the earth and the stationary place of the Polar star.

What appearance would the heavenly bodies present, were the earth like a big drum rolling over?

Why, I should fancy that then they would all rise in the east and pass over, in an orderly, proper manner, to the west. Though I have been told that the earth is round I

do not understand how that should affect the motion of all the stars, as we see them.

Come, let us see. If the world be round, then some parts would describe a less circle than others in the daily revolution.

Of course, father.

Would there be any portion that would not turn at all?

Yes; the poles, or extremities.

Where would be the most motion?

At the equator.

Then, if the stars be still, and the earth move and be round, would not--

O, I know all about it now. The stars over the pole would appear still, because the pole is still. The stars over England would appear to describe a larger circle than those of the Great Bear over Sweden. Where we are now, the stars overhead describe a very large circle.

And at the equator the greatest circle of all?

Yes, yes,--and I can now see why the Great Bear seems to go away from us, as we get up the hill toward the equator. That constellation is opposite to a part of the earth a good way to the north of us. As we rise higher from that place, we must appear to rise higher than the stars over it. How I do long to stand under that bright Belt of Orion, and under Sirius! And how I should like to see the pretty Cross stretched over the mast-head!

Wait a bit, my lad, and you may see all these wonders in time.

How delighted I am to make out that all those funny motions among the stars are simply owing to the world being round."

SIZE OF THE WORLD

It was while half tired of ship life already, and longing for the excitements of shore in a new country, that the boy turned round to his father, and said, with a sort of yawn--

"What a long while we shall be getting to Australia, father.

It is a great distance to go, James.

Well, I think the earth must be a good big ball to roll about among the stars.

You ought to know its size.

I have been told that it is 25,000 miles round; but who has measured it.

Why as to that, you and I might measure it some evening by the stars. You know that the great space between the Belt and Polar Star is one-quarter of the great circle of the heavens.

Of course.

Every circle is reckoned to be divided into spaces called degrees, of which 360 go to form the circumference; so that one-fourth will be ninety degrees. You can imagine that space divided into ninety of these portions, called degrees.

What rare compasses that would take!

Those two bright stars overhead, which are about twice the same distance apart as the apparent diameter of the Sun, would be nearly equal to one of these ninety portions. Now, you know that if you were at the pole, you would have the Polar Star above you; and, if at the equator, Orion’s Belt would be over your head.

I understand all that, father.

You know, then, that you might be in a place where one of those two bright stars would be overhead, and you might journey on to another place further south, where the other one would be at the zenith or overhead.

This is all clear to me.

Then if you measured the ground you had gone over, it would be equal to one-ninetieth part of the space between the pole and the equator, or about seventy miles. Can you tell how many miles it would be from the pole to the equator?

That would be seventy times ninety--which is 6300.

But that is only one-quarter of the way round--is it not?

Yes; the whole distance round would be four times 6300, or about 25,000 miles.

Well done--we have measured the circumference of the world.”

THE EARTH NOT QUITE ROUND.

“Father,” exclaimed James one day, “the captain has been giving me a riddle which I cannot guess. He asks where a pound of shot is not a pound. Is he joking with me?

Not at all, lad. A bag of shot weighing a pound in London, or even in Melbourne, will not weigh a pound on the equator.

What queer things you do tell me! How can that be?

You know that the world attracts you.

Yes; I know when I jump up I am soon brought down, and sometimes unpleasantly so, too.

As the world is round, and everywhere the world pulls things down to its surface, it is as if it were drawing all to the centre. This is called the _Attraction of Gravitation_.

And I suppose that like as fire is hotter the nearer you get to it, so does the attraction get greater the nearer you get to the centre of the earth.

True; but can you guess how this gravitation can be weight?

I know when I pull a thing, it is as if I pushed it, or put a weight on it. So the pulling of the earth will make the thing attracted feel as if a weight were on it.

And can you not see why a pound should somewhere not be a pound?

No; unless I made a big hole towards the centre, and then the pound weight would be heavier. But you told me the world was round, and it must be as far from the centre at London and Melbourne, as at the equator.

But if from trial it be found that the pound really is less at the Line, what would it prove?

It proves what I can't see, that the earth at the equator is farther off the centre than what it is at London or Melbourne, and that would make the earth not a true Globe.

Neither is it my son: though the difference is but little. It is about a dozen miles thicker down to the centre at the Line than at the poles.

Then, as London and Melbourne are each nearly half-way between the line and the pole they will be nearer by six miles to the centre than the equator is.

You have solved the problem, and understand that the world is not a sphere, but what is called an Oblate Spheroid, being flattened at the poles.

As I cannot easily find out by weighing that a pound is not a pound: can you tell me any other way of proving that the earth is not quite round, dear father?

Yes, I can. Do you see the reason why the pendulum returns down when it is moved up?

The gravitation, to be sure.

If a thing weighs heavier in one place than another through a difference of gravitation will there be a difference in the two places with the pendulum?

I should fancy that where the earth pulled the harder the pendulum would come down the quicker.

If it came down quicker it would rise quicker; that is, the beat would be quicker. What effect would this have upon the clock?

The faster it beat, the faster the wheels would be moved, and the faster the hands would turn on the face.

Just so: the clock would be faster in time when the pendulum beat quicker, and slower in time when the pendulum beat slower.

But how could I find this out, father?

This way. It is ascertained that in London a pendulum must be 39-1/7 inches from the top to the middle of the swinging-bob to beat a true second at a time.

Then I am sure the London pendulum will not be pulled down so hard at the equator, and so it will be longer coming and going.

What must I do, then, to make my London pendulum beat seconds on the equator?

Why, shorten it a little bit.

Yes; or, what is the same thing, move up the weight at the end of the pendulum a little. I must shorten it one-tenth of an inch, or else it will lose sixteen hours in the year.

Then all the clocks carried to Melbourne from London will be wrong.

Will they go too fast or too slowly in Victoria?

They will be a little too slow, as Melbourne is nearer the equator than London is.

If a Melbourne clock-maker want his second pendulum to beat seconds in London what directions must he give?

He must tell the folks in London to lengthen the pendulum a bit.

You will see that by the world being flattened at the poles the degrees of latitude will differ. One close to the equator measures 68-3/4 miles, in Victoria about 69 miles, in England 69-1/8, while in Lapland it would be 69-1/4.

What would be the mean length of a degree?

About 69-1/20 miles.

Let me multiply that by 360, to find the mean circumference of the world; I have it--24,858 miles."

ANNUAL MOTION.

The ship was much delayed in the Region of the Calms, as it is called; that is, near the equator.

After a while the astronomical dialogue was resumed.

"Well, James, look now at your twinkling friends, and tell where you are.

I suppose by the North Polar star resting upon the waters to the north, that we must be now nearly on the top of the hill, by the equator.

We are so, my lad. But what time did the sun rise this morning?

At six, and it set at exactly six this evening.

Do you know what day of the month it is?

The twenty-third of September.

Have you ever observed the sun to rise and set at six o'clock before, on the voyage?

No, father.

Well, then, you will recollect hereafter that you were upon the equator on the twenty-third of September, when there was equal day and night. But did you not know of equal day and night at home?

Yes; about half-way between winter and summer.

Just so; that is, at the end of September.

Why should it be so there in England and here too?

We will try and find out. I think I have pointed out to you the Pleiades.

I know them well. They are in the Bull cluster.

There happens to be a band of twelve constellations forming a broad circle of the heavens.

Is that what is called the Signs of the Zodiac?

The same. Can you repeat them?

Yes, father. Aries the Ram, Taurus the Bull, Gemini the Twins, Cancer the Crab, Leo the Lion, Virgo the Virgin, Libra the Scales, Scorpio the Scorpion, Sagittarius the Archer, Capricornus the Goat, Aquarius the Water-bearer, Pisces the Fishes.

Over there to the eastward is one of these that rose as the sun was setting. The sun is seen opposite one or other of the twelve constellations in each month of the year. It is observed, for instance, to rise each morning for a month in the constellation of the Ram, and for the next month in the Bull.

How can it be among those stars when you say they are so much farther off?

By saying it is in such a constellation, I mean that it is in a line between us and that cluster of stars.

Why, then, the sun appears to go a journey between the stars and us for a whole year. Then it will rise in the Ram twelve months after it had been seen opposite that constellation. I wonder how this can be. Did you not say, father, that neither the stars nor the sun really moved?

I did.

Well, then, it must be either the stars turning about the sun, or the sun taking an annual trip about them, and yet the stars do not move, and the sun does not.

Let us leave the sun and his trip for a little, while we have another talk about the stars.

Notice that red star just rising there on the larboard side.

I have it, father.

As I want you to remember it, take my pocket-book and mark off with the pencil a picture of the star and its four neighbours, putting a cross under the one that has just risen.

There it is,--the under one of this cluster I have sketched. I shall know him again.

Now, then, you must go below, and resume your observations to-morrow. I took the exact time--seven p.m.--when you saw the star rise. You must look for it at seven to-morrow evening."

At four minutes before seven the next night, up rose the pretty object, and stole a glance at the boy across the ocean.

“Father, father! it’s up, it’s up!” sang out the happy lad.

His father looked at his watch, and said--

“Your star did not come to his time. It was four minutes to seven o’clock when he appeared, and that is four minutes too soon.

How is that, father? I must ask you, for it is of no use asking the star.

Perhaps you will still wonder when I tell you that to-morrow evening it will be eight minutes to seven before it gets up at this place.

Why, this is like the sun among the stars--always getting out of his place. But the sun gets right after a year.

Do you think our star will ever get right--so as to rise at seven some other evening?

I do not know, father.

If it rise four minutes earlier every day when will it be an hour earlier?

In half-a-month; and twenty-four hours earlier in twelve months. But that is the time when it will be visiting at night again.

So you have ascertained that the stars go an apparent journey round the earth in a year; like as the sun takes a year to go his apparent course among the stars.

Then both must have the same cause. As I found the daily motion of the stars was owing to the real daily motion of the earth, I suppose this yearly motion of the stars must be caused by some yearly motion of the earth.

Either the sun turns round the earth once a year, or the earth does round the sun.

Well, father, the sun appeared also to turn round the earth once a day. I do not think I could run round the deck in two minutes while I am two hours doing it. Then, as I am satisfied it did not have a daily motion, I am the more disposed not to trust to this apparent yearly motion. It is more likely, too, that the little world should run round the great sun, than that the great sun should revolve round the earth.

I perceive you are as settled upon the annual or yearly motion as you were upon the diurnal or daily revolution.”

THE SEASONS.

“James, my boy,” said Mr. Marple “do you know why there should be winter and summer?

No, father, I really cannot tell. I know that in England the sun was higher up in the day, and that he stopped up longer in summer than in winter. But then as I believed that the sun ran about as he liked, I was not surprised at his change of place. Now, however, I am very anxious to know more about this subject.

When were we in the circle of the Pleiades, that is, the place where at some time or other of the twenty-four hours they would be overhead?

O, that was long before we got to the equator.

Yes, it was about 1500 miles to the north of us. Now we are in the circle of the Virgin cluster.

But the Bull and the Virgin belong to the twelve constellations, of which we were talking, forming a broad circle, along which the sun appears to travel.

True. But they are not in an upright circle. Their line will cross our equator Line.

If that be it, the sun does not appear, then, to go round us over the equator.

No; for if it did, it would be seen overhead at that place every day at twelve o'clock; whereas it shifts its course continually. One day it is over one part at noon, and upon another day over another part.

Why, if the circle of the Zodiac be inclined to the circle of the equator, it will cross it in two places. I suppose that is the reason why, on the celestial globe, one constellation, as the Virgin, is over the equator circle; others, as the Bull, over some place to the north of it; and some others, as the Scorpion, to the south.

You guess rightly.

Then if that be it, the sun must appear to cross the equator twice in the year.

It does so on the 21st of March and 23rd September. These are the vernal or spring equinox, and the autumnal equinox. March is the vernal equinox of London, or equal day and night in spring; but it is the Australian autumnal equinox. In September the Australias have the spring equinox, and the English the autumnal. The sun is seen overhead to the furthest north on the 21st of June, and to the furthest south on the 21st of December.

Why, the first is the summer of England, and the last the winter. Then I suppose that is the reason why in summer the sun looks so high up at noon. Then in winter, as it is shining more over parts south of the equator, the north would neither see nor feel so much of him.

That is correct. You now see why the Australian summer is at the time of the English winter.

Yes, father; because they are on opposite sides of the equator. But you know that the sun does not really move about like this. I want to know what does move.

The earth, to be sure.

I understand that; but neither the rotundity nor motion seems to explain this mystery of the Zodiac. But let me see. When the world swings round the master light, is it obliged to keep bolt upright--north pole up and south pole down?

How would your equator then be, James?

Right opposite the sun. But that will not do, for it is not always opposite to the sun. Perhaps the world goes waddling along like a duck, first on one side a little bit, and then on the other. This would bring the sun opposite to either place.

It would be a very regular waddle. We will try your scheme. Pick up that round wooden ball there. Carry it round the capstan, and see whether by twisting it from

one side to the other you can keep the south pole shone upon for six months, and the north pole shone upon for the other six months."

James now makes a trial. But after all sorts of turns he failed in producing the English winter and summer in the proper time and for the proper length of time.

"I must give it up, father.

Just slant your ball a little out of the perpendicular, and carry it round your sun without moving it from its position.

Capital! It explains the whole. When it was in front of the capstan, the south pole was toward it; that was the Australian summer. When I went to the right side, neither end was inclined toward it; this was the sun over the equator. When I came to the back of it, the north pole was toward it; this was the English summer. When I reached the left side, my equator was opposite to it; this was the sun's second crossing of the Line. When I got to the front again, there was my south pole pointing to the capstan again.

Then you really understand the seasons. Of course the spring in England is the autumn of Australia. But I have something else to recall to your recollection. Have you forgotten the old church on the hill at home?

No, father, and never shall. How pretty it looked when the sun rose over it upon a fine spring morning, with the birds all alive with their song, and the bees down upon the flowers!

Did the sun rise over the church at hay harvest?

No; I remember then it came up from behind the squire's house to the left of it.

Where was the sun first seen upon a cold winter's morning?

Aye, I fancy I see his red face looking lazily over the parsonage to the right of the church, and that when we had done breakfast, too.

How came the sun to jump about in that manner?

Why, do you see, he did not hop from one side to the other. He would get a little further from the church each morning till he rose over the squire's house: then, as if ashamed, he seemed to creep back to his old place. Afterwards he appeared to take it into his head gradually to get on to the parsonage, and then he would once more turn to the church.

Which was on the north-side of the church, James?

The squire's house. Now, I think I can explain all without your questions this time, my dear father.

Then go on by all means.

The sun rose over the squire's house in summer, for that was the north side; the north pole was then inclined to the sun. At spring it was half-way between the squire's and the parsonage; this was the sun over the equator. In winter the sun rose at the furthest part to the south; this was where the north pole was turned away, and

the south side turned to the sun.

Well, then, you have satisfactorily explained the seasons."

PRECESSION OF THE EQUINOXES.

It was with no little impatience that James had waited for the first appearance of the Southern Cross, the emblem of Australia. He first noticed it when a long way north of the equator, and each evening afterwards bestowed a considerable attention upon it. As he made southing on his voyage the glorious constellation rose higher and higher in the heavens. Hour after hour would he sit, marking its progress, wondering much to see it more or less upright according to the time of night, and more or less declining from the perpendicular at the same returning hour of evening upon different days.

James could understand the Cross seeming to perform a circle round the south pole of the heavens, as he had seen the Bear round the north pole, from the rotundity of the earth and its daily motion.

It was quite natural, then, that the lad should be so enthusiastic when he spoke of the constellation to his father in these terms:

"O that pretty Southern Cross! I like it more and more as I get down to the south. I did feel sad to see the old Bear drop gradually, night after night, till it seemed to sink for ever in the northern Atlantic; but I have a new friend here, that keeps rising higher and higher each night, as if to welcome us to our southern home. How it would please many boys in Europe to see the Cross!

But it was seen in Europe formerly.

Another puzzle, dear father. Did it run away from the south pole to have a look at the north one?

It was not a run, but a gradual slide out of its place, and as gradual a slide back into it again.

What a queer trick! Is it on its rambles now, I wonder?

It is always on the move.

Yes; I know it describes a circle in the sky once in twenty-four hours, but that is owing to the world rolling over to get its daily light.

Yet it has another motion; not really its own any more than the other. And this leads it further away from the central point over our south pole, and then brings it to its place again.

Well, I am glad it is here now for me. Will there be time for my London cousins to catch a glance at it when it wanders northward again?

O no, no; it does not move quite so fast as that. But if Abraham had been sojourning in southern Europe, instead of Asia, he could have seen it, though it was then making its way back to the south.

How could the Cross slip away from the other southern stars to go on such migrations?

Not so. All the stars keep their places relatively to each other, as you see them do in the nightly progress from east to west. In the daily motion none get before the other, nor did the Cross get before its neighbours.

Well, I am fairly done. That is a riddle.

Look at the question, boy. What makes the apparent daily motion of the stars?

The real daily motion of the earth.

And if, then, you observe any other peculiarity of movement among your bright friends up there, to what may you reasonably ascribe it?

I should imagine some peculiar twist, roll, or slipping of this world of ours.

True. And if there be seen among the polar stars, north as well as south, a slight but regular movement, a sort of a swing round, so that some stars get farther off the spot we call the south pole of the heavens, while others approached it nearer, and yet so swung round that at last all find their old places again, how could you get a motion of the earth to make up for all that?

Ah! I must contrive that the world should do three things. It must roll over once a day, roll round the sun once a year, and yet wriggle about in another way at the same time.--Stop a bit--I have it. My top turns round itself, and pretty quickly, too; it shifts about on the floor almost in a circle as it turns round; and I see it, especially when it is quietly spinning, have a slow roll of the head, like a sailor when walking ashore, as if its head were a little giddy.

You have hit it exactly, my good fellow. The three motions of your top are much like the three motions of the world. That top-heavy slow swinging of your top while it is spinning in full force is like a sort of head-rolling of the earth. The poles seem to have a roll of their own, independent of the regular roll.

Yes; and that would make the Polar stars seem as though they were swinging. But how long are they before they are in their regular places again?

Astronomers calculate 25,850 years.

But will not this changing make the star-charts of the ancients all wrong?

Indeed it does. Aries the Ram, for instance, as a _sign_, is the first thirty degrees on the celestial globe; but as a _constellation_ it has shifted to between the thirtieth and sixtieth degree on the ecliptic. This is called the precession of the Equinoxes.

How is that? I know when the equinoxes are--March 21st and September 23rd--when it is equal night all over the world.

The European vernal equinox took place at the first point of Aries. But by this precession, or more properly recession, or going backward of the stars, the Equinox takes place when the sun is in Taurus the Bull, two thousand years after it was in the Ram.

I see. The stars not only get earlier four minutes a-day to accommodate the earth in its annual motion, but make a change to accommodate the swinging of its pole.

Can you tell me, James, how much the stars slip back in a year?

Let me calculate it. There are 360 degrees in a circle, and sixty times as many minutes; that is, 21,600. As it is 25,000 years in the circle, the stars would not shift one minute of distance, and the sun's apparent size in the heavens is thirty minutes, or half-a-degree.

By this Precession a star comes to the same spot about twenty minutes later every year. It is, as it were, sliding back. In Europe, James, one effect has been to bring what we call the Polar star within one degree of the north pole of the heavens; whereas, a few thousand years ago it was twelve degrees off it. The bright star of the Lyre Constellation will some day be the North Polar star.

Ah, but in 25,000 years the present Polar star will be in its old place. The end of the little Bear's tail seems now fastened on the north pole; but it must be a sort of greasy pole for the stars to slip off it as soon as they get near it."

FIXED STARS NOT FIXED.

It had been a sultry day. The sun rose with a fiery glance, and the sea blushed with a deep red as his hot beams glided over the heaving surface. The clouds hung heavily and gloomy in one part of the horizon, sending back to the sun a glowing, angry look in exchange for his fierce gaze. It was not long before the massive clouds came rolling up, as they were pressed forward by the western breeze, which was rushing forward as hastily as the rest. The sun tried to frighten the advancing vapours, but could not. They came onward till they covered the sky, and drove the bright rays from the gilding of the waters.

The wind, rejoicing in its power to drive the clouds, now flew upon the ocean and buffeted the waves. Affrighted at its blows, they fled hither and thither in their distress, and mingled their roar of terror with the shrill shrieks and cries of their invisible foe.

It was a storm. The captain, who had been attentively watching the signs, and losing his breakfast, saw what was coming, and prepared for it. The sails were taken in, and the ship made snug and safe. The storm came with a wild burst all at once, and reeled the vessel almost upon its side. But the officer was there, the helmsman was awake to his duty, and the lounging barque kept on her course in spite of her erratic movements.

James was not insensible to fear. He was but a boy. And when, in addition to the noise of wind and billows, the thunders came pealing from the heavens, he instinctively clapped his fingers to his ears, as if to drown the sound and his fears. The vivid stream of lightning, as it darted from cloud to cloud, played round the ship, or plunged madly into the sea, added to the interest and the terror.

But the echoes died away to whispers. The clouds put on their snowy robes. The ripples gently laughed on the bosom of the ocean. The sun, no longer angry, smiled

kindly upon the scene. In the evening, a few lines of vapour were motionless in the sky; and the long, uncertain heavings of the waters told that a calm was softly ruling all.

The boy was found by his father looking at the placid face of the heavens.

“What, boy! at your old post, staring at Orion again.

I’ll tell you what I was thinking of, father. You see the world is like this ship--always on the rock, never really still--and I was thinking how pleasant rest is, and what a lot of it they must get up there in the fixed stars.

Not so; we are now learning that they are, like some boys, not so steady as they look.

You surely don’t mean to say that such respectable old folks as Sirius and his bright friends ever go dancing about in the heavens like giddy Venus and Mercury? I know they appear to tumble about over our masts in a storm at night, but that is because the ship is rolled over by the winds and waves.

It is really true that they do not keep to their places, though the change of position is so very slight as not easily to be discovered. Your friend Sirius, for instance, has been closely watched, Dog as he is, for two thousand years. We know what his place was that time ago, and what it is now; and it has shifted to the southward half-a-degree.

That is as much as the apparent diameter of the sun. But has he any company in his rambles?

Yes; the _Bull’s_ Eye, Aldebaran, has kept up pace with him.

Quite right to have an Eye upon the Dog. But has any exact difference of position been observed since we have had good telescopes?

The star sixty-nine of the _Swan_ has been well watched for fifty years, and found to have gone on four minutes--the eighth part of the moon’s face. A star in the _Indian_ has moved seven seconds in the year. They are not fixed, like the ancients thought they were, in a crystalline sphere; or riveted, as Aristotle taught.

Pray, is our sun no more fixed than the rest of them?

He, too, is on the move.

But where are they all going to?

That has been a matter of dispute. Some thought the stars were dancing round Sirius. But Mädler, the German astronomer, would have us believe that they have a greater fancy for some spot near the _Pleiades_. Others think of Hercules. But all would take many millions of years.”

DISTANCES AND SIZES OF THE STARS.

James had a long talk with the captain one day about the telescope, and got a capital lesson about the magnifying power being according to the character and size of the glasses used. The sailors called the instrument the “Bring ’em near;” and it

does make distant objects look as if near. It was explained to the boy that the stars were of different magnitudes or sizes, first, second, third, &c., according to their relative distances. The following conversation followed between the father and son:--

“Now, my boy, I must try and give you some idea of the penetrating power of light, that you may get an idea of the enormous distances of the stars. You are aware that the more distant they are the less their light. A star of the first or nearest magnitude will have one hundred times the light of one of the sixth magnitude. A telescope, therefore, gathering one hundred times the ordinary light, will make the sixth look as near as the first.

And will it be one hundred times further off?

No; light increases or diminishes according to the square of the distance.

I know. If the light be one hundred times less, the star will be ten times further off, for the square of ten is one hundred. I can understand now that a thing is only seen by the help of light. I do not see many stars, because their light is too little for my eyes to take in. The telescope has bigger eyes to take in the light of the distant stars and nebulæ.

The pupil of the eye is but one-eighth of an inch in diameter. An object glass of twelve inches diameter is, therefore, ninety-six, or say one-hundred times as long. As the light seen is according to the square of the diameter, the telescope of twelve inches will receive one hundred times one hundred, or ten thousand times more light.

But is there a way of measuring the quantity of light?

There is. We find that the sun has twenty-two thousand millions of times more light than the nearest of the fixed stars.

Then, the square root of this ought to tell how much further off it is. Let me see. It will be about 150,000.

Yes. If the sun were put back 150,000 times further than it is, it ought to look as brightly as that star. If it does not, it is because it is really smaller than the star.

What! 150,000 times 95 millions!

But that is nothing; for it is only to the first rank. What of the twentieth magnitude?

Yes. But you say the nebulæ are further off than that.

I may tell you that if the sun moved three times as fast as the world does, in its six hundred millions of miles a year, it would take two hundred and fifty millions of years to get to as far as Lord Rosse’s telescope could see.

That takes my breath away.

Hear a little more. Light comes from the sun to us in eight minutes. It will take sixty thousand years to come from one of those stars Lord de Rosse saw. In fact his telescope has enlarged our universe one hundred and twenty-five million times.

Then I think religious people ought to thank astronomers for showing them more of the greatness of God. Those who only thought of him as the Creator of the three

thousand stars, to be seen by the naked eye, could not have such a notion of his vast power as those who know of millions upon millions of suns.”

* * * * *

After this, James was left to digest such wonderful lessons. When his first astonishment had passed away, his curiosity was excited to know more about the distances of the stars, so that he might form a simpler idea of the thing. He took, therefore, another occasion of bringing up the subject in these words:--

“Father, do you really believe the stars are so far off?

I am obliged to believe many things I do not understand, upon the testimony of trustworthy witnesses; but in this case I can form a good guess of the truth. Do you remember what I once told you of the parallax, or angle of observation of the sun or moon?

Yes. That of the moon was 57 minutes, and the sun 8-3/4 seconds. A degree is 60 minutes, and a minute 60 seconds.

Very well. Then 57 minutes, or 3420 seconds, will be four hundred times as much as the other. If the moon be 240,000 miles off, the sun will be four hundred times further or 96,000,000.

But how do you get this parallax?

Distances are calculated by the angle made in looking at an object from two places. The two lines of sight cross one another. A great base is needed to view a distant object, or else no angle can be observed. Astronomers take the diameter of the earth’s orbit.

That is twice ninety-five millions of miles.

With that base--that is, looking at a star from both sides of our orbit, or at six months’ interval--we could get the two lines crossing one another, and so making an angle. The further the object, the more minute the angle. Only a few of the fixed stars could be observed in this way, as they generally are too far off to give an angle.

I know an equilateral triangle has its three angles equal to two right angles; and with ninety degrees for one right angle, each angle of the triangle will have sixty degrees. But I suppose no star parallax could be one degree.

No; nor a minute, the sixtieth part of one degree. When the object makes an angle of a second, or sixtieth of a minute, from a base line of one hundred and ninety millions of miles, the distance of the star will be about twenty millions of millions of miles.

Is there any star making the second angle?

The _alpha_ of the Centaur is about that, and is one of the nearest of fixed stars.

That the nearest to us, and yet so far! Do tell me the distance of some others.

There is one, 61 Cygni, of the Swan, with one-third of a second; and, therefore, three times the distance of the _alpha_ Centaur. There is a star in the Lyre which is

one-fifth of a second. Grand Arcturus is one-eight. The North Polar Star is one-tenth. Pretty Capella, of the Kid, is one-twentieth; that is, twenty times farther back than _a_ Centaur. As it looks one of the brightest stars, it must be very large.

What of old Sirius?

The angle he makes with our orbit diameter is one-fourth of a second; so that he is about 80,000,000,000,000 miles.

Thank you, dear father, for these terrible long figures.

Their great distance may give us a good guess of their great size.

I know the size of the sun to be half-a-degree in the heavens, at a distance of ninety-five millions of miles, and yet it is really eight hundred and eighty-six thousand miles diameter, I have been told. When, then, a star is millions of millions of miles away, I am sure it must be a big one to be seen at all.

It is so. If the sun were thrust back as far as the star _a_ Centaur, it is calculated that it would not shine with more than one-third of the light that star now gives us; it must, therefore, be not more than one-third its size.

But you said Sirius was four times farther off than the Centaur, father.

Yes; and it gives four times as much light. It is, then, probably four times as large. It must be, therefore, many times larger than the sun.

I am sure it ought to have more planets turning round it than the sun has, or else have them much larger in size."

CONSTELLATIONS.

It was a calm, still, moonless night, which closed in with raven wings a glorious sunset in the South Atlantic ocean. The gay clouds, in brilliant dress, that hung about the sun as he glowed to his evening rest, had gradually changed their garments for those of more sombre hues; and then, as gradually, had changed from mountain shapes to feathery forms, till softening night dissolved them into invisible vapour.

The stars came stealthily forth; a few bolder ones rushing first to the front, while the more modest stars feared to show themselves till the last sign of daylight fled.

First, second, third, fourth, and other magnitudes thus successively gleamed from the cloudless arch, above and around the boy.

After a time he said--

"Why were the Ram, Bull, Lion, Scorpion and others so called? I have often wondered at their odd names, and have tried to make out a bear in the Great Bear, or a dog about the stars near Sirius, but never could see any resemblance to the animals.

I don't wonder at your surprise. Some learned astronomers of our day pretend that the ancients by the Nile and Euphrates had no meaning in the names they gave to the constellations. This is not likely, especially as we know that the signs of the Zodiac were pictured in their temples.

But who gave the names to the other stars that are not signs of the Zodiac?

The ancients. But it is time now to sleep away all thought of the stars in the care of Him who cares for the stars."

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During the night our young friend had a disturbed rest. The constellations kept circling round him; while lions, bulls, fish and crabs were mingled, in strange confusion. They glared at him with their hollow eyes till the boy was quite frightened, and cried out in his sleep.

The morning sun awoke the boy, and put to flight the stars. When sweet evening came the twinkling lights stole forth again, and the astronomical dialogue was renewed by James.

"Please, father, I want to know the various constellations.

We will class them, my son, into North Polar, South Polar, and Equatorial.

The North Polar star, at the end of the Little Bear's tail, is a capital guide for the north pole; but I don't see any Little Bear southward, walking round his tail-tip, like a horse tethered in a paddock, so that I may discover the south pole.

No, we have not a South Polar star, and no bright stars at all near the south pole of the heavens.

Let us have a little more about the Zodiac first, father. I know the twelve signs are constellations over the tropics; but I don't know why you call them the Zodiac.

Zodiac is from a Greek word, meaning a small figure of an animal, as they are nearly all animal figures.

How many northern and southern constellations are there?

In the ancient celestial atlases we find twenty-one northern and fifteen southern, besides the Zodiac; then more were added of new ones named or old ones subdivided, till we have, including the Zodiac, sixty-two northern and fifty-five southern constellations.

Of course, the old astronomers of Egypt and Babylon could not have seen all the south polar stars?

We should hardly fancy it. Strange tricks have been played with the names of the constellations. Only think of English writers, sixty years ago, wanting to have your Orion called Nelson; and the University of Leipsic, to please the French emperor, proposed to steal the belt and sword from Orion, and call them the constellation of Napoleon.

I am glad they didn't, though. But how is the line drawn to distinguish the spaces of the constellations?

Their sizes vary. Orion, for instance, takes up a great space in the heavens, while others occupy little room.

How many stars go to make up a constellation?

You mean those seen by the naked eye. These differ in number. I will run over some of the Zodiac. Aries has 66, Taurus 141, Gemini 83, Cancer 85, Leo 95, Virgo 110, Libra 51, Scorpio 44, Sagittarius 69, Capricornus 51, Aquarius 108, Pisces 113.

But there must be a lot more seen in them by the telescope.

Yes. The six stars of the Pleiades turn out to be 188. In Orion there are 2000 stars easily distinguished. Your three in the belt are 80, and the star in the sword you find to be a dozen.

Will you kindly show me, dear father, how I can find the south pole of the heavens, as there is no bright star near it?

I will try and give you a rough chart. Look at the Southern Cross. Draw an imaginary line from the top star through the one at the foot; and four times that distance in the same line will bring you to the south pole of the heavens.

Yes; and if I had that point over my head I should be where no one else has been--at the south pole of the earth. Now I have my centre, I would like to describe a circle at the distance of the Cross and know the stars inside of it.

Begin, then. Look at the two bright ones pointing toward the Cross beside it. They are the Pointers to the Cross. A lot of bright stars together, on the other side of the Cross, are in the Ship, Argo. Follow on the circumference that way, and you arrive at a large star, a little north of our course, called Canopus. From the Cross to Canopus is one-quarter of the circle round the pole, and it is filled up with the Ship.

What a ship!

It goes northward, too, to Sirius, the Dog-star. Canopus is half-way between Sirius and the south pole, and is in the line between the two.

That is a capital point, for I know them.

South of the Ship, between it and the pole, are the little Flying Fish. The Table Mountain and Chameleon, with feeble stars, are between the Fish, again, and the poles.

Now I have that quarter of the circle, well.

Keep on to the right, still away from the Cross and beyond Canopus, nearly as far again, and you see a bright star in our circle, which is called Achernar, at the bottom of the long River Po.

Not the Italian river, but the Celestial one.

That heavenly stream runs northward to Orion. Now, between Canopus and Achernar we see but few stars, although we pass the constellations of the Painter’s Easel, Rhomboidal Net, and Clock. Between Achernar and the Pole are the tiny stars of the Little Hydra. But between the Little Hydra and the Flying Fish, resting like on the Table Mountain, is the beautiful Nubecular Major, the great Magellanic Cloud.

Now I know where I am again. The Rhomboidal Net is on one side of the Cloud, and the Flying Fish on the other. The Table by the Pole is south of it. But what is north of it, for Canopus is not quite in a line with the Cloud and the Pole?

The Sword Fish is on the Cloud. But go again to Achernar, and draw a line thence to the Pole.

Why, it goes right through the Little Magellanic Cloud.

That Cloud is between the Hydra and the Toucan. That bird's head beyond Achernar is in our circumference of the Cross. A little further on that round line is the bright star in the Peacock's head; though the Indian, with his arrows, stands between the Peacock and the Toucan.

The very place for a hunter. But go on, please, for you are fast completing the circle, by getting back to the Cross.

The end of the Peacock's tail brings us three-fourths round; and above it, from the Pole, is the Southern Crown.

I see it; but it ought to hang over something better than a Peacock's tail.

It so happens that the Telescope comes between them. Below the Peacock, and close round the south pole, is the constellation Octant. Now for the last quarter, between the Cross and the Peacock.

I know part of that--the Cross's Pointers. But what are those three stars a little below the Pointers, toward the Peacock?

They are the Southern Triangle. Between the Cross and the Pole is the Fly; while between the Triangle and the Pole is the Bird of Paradise.

What a cold place to put so delicate a bird! But to what constellation do the two Pointers to the Cross belong?

To the Centaur's two fore feet. The last of my circumpolar stars are those of the Altar, lying north of the Triangle, a little to the left of the Pointers. You may see them on the edge of the Milky Way--two stars close together, and four others to the right further apart. The Compasses, hardly to be noticed, are between the Altar and the Pointers."

* * * * *

This was a long lesson, requiring much study afterwards. When Mr. Marple thought the lad knew the southern stars near the pole, he was ready to advance with the constellations again, and proposed to take a fresh round.

"We will," said he, "take another sweep round the heavens, embracing the space between our old circle of the Cross and that over the equator, though I will not take equatorial stars now.

May we begin at the Crown, father?

Between the Cross and Crown is the Altar. North of and near the Crown is Sagittarius the Archer, beside which is the Scorpion.

How can I tell that?

The Scorpion has a line of bright stars coming down south to the Altar, with a twist like a curl in its tail.

What is that bright red star at the upper end of it?

Antares of the Scorpion. The bright star near it further north is in the Libra or Scales. The fine star as far from the Pointers as they are from the Pole is in the shoulder of the Centaur, half horse and half man.

Is he fighting anything?

Yes; he is running his spear through the mouth of the Wolf, which is stretched between Antares and the Pointers. The Centaur comes down to the Cross, and goes up north to the great Hydra, on which the Crow is perched.

What a beautiful star that is over Centaur, about twice as far from the Cross as that from the Pole!

That is Spica in the Virgin, close to the Crow. That makes up one quarter of our second circle. Looking still to the right, and north of the Cross, do you see any good-sized stars?

No, father, there is nothing worth looking at for the next quarter beyond the Cross and Ship till we get to Sirius.

Yet there are the Pneumatic Pump and the Mariner’s Compass. The great Hydra stretches itself all from the Scorpion, under the Crow, the Cup, and the Sextant, from near Antares to Canis Minor, the Little Dog.

Where is the Little Dog?

To understand that we must go on to our third quarter of the circle. You know Sirius and the three starred belt of Orion. The Belt is exactly over the equator. The three stars lead you down south to Sirius, and a line at right angles brings you to Procyon of the Little Dog, which thus forms a triangle with Sirius and the Belt. Get the Pole, Canopus, and Sirius in a line, and a little further to the north, on the one side you have the Belt, and on the other Procyon.

Though I see no stars of any consequence between the Great and Little Dogs, is there any constellation there?

Monoceros, the Unicorn, keeps them apart. The Hare and Noah’s Dove are on the other side of Canis Major. They are both south of Orion. In the Dove, a little out of the line half-way from Canopus to Sirius, is the pretty Phact. The five stars in a line between Canopus and Sirius are in Canis Major, the Great Dog Constellation.

What a gay show of stars between the Pole and Orion! But what a blank beyond to the right. I see one, however, opposite to Sirius, forming a parallelogram with it, the Belt, and Phact.

That is Zaurack, in the River Po; and the great space beyond that, to the right, far north of Achernar, is Cetus the Whale. The star Menkar or Menhab, a little north of

Zaurack, is in the Whale's head, while Diphda is in its tail. Achernar is about half-way from Diphda to the Pole. Between the two are the Chemical Furnace, the Sculptor, and the Phoenix.

I don't care about these. Have you not nearly got round, father?

The last quarter of our circle lies between the Southern Crown and the Whale's tail.

That will be north of the Peacock and Indian. What is the bright star between the Whale's tail and Achernar?

The head of the Phoenix. And forming a triangle with it and the tail is the noble Fomalhaut of the Southern Fish, under the sign Pisces and near the toe of Aquarius. The Goat Capricornus is to the right of Fomalhaut.

Why, there is nothing hardly between Fomalhaut and the Crown.

Between the Peacock and the Crown are the two stars close together, forming the foot of Sagittarius, whose body stretches from the Crown to Capricornus; while the Crane and the Microscope lie between the Peacock and Fomalhaut.

It would require a good telescope to see that faint-looking Microscope.

You will see seven stars in a line, left of the Peacock, dipping down toward the Pole--these form the Crane.

And a long-necked fellow it is, too."

* * * * *

Another rest was necessary here, to enable James to connect in his mind the last collection of constellations with his circumpolar ones. His father then took him to those about the equator and north of it, and thus introduced him to the twelve signs of the Zodiac.

"You have learned, my dear boy, the names of these signs; I want now to teach you their positions.

Yes; I would like to know how big the Bull is. Do his horns poke the polar star, while he hangs his tail over the equator?

Our Bull is a funny one, for he has no tail, because he has no hind quarters. If he reached from the pole to the equator, a few Bulls would fill up the sky, and leave no room for Bears, Swans, Hunters, and other creatures up there. Now look at the Orion's Belt, and run down the line of the three stars as you did to Sirius, but in the opposite direction, as far north as the other is south.

I have it--a beautiful star between the Belt and the pretty Pleiades.

That is Aldebaran, the Bull's eye. Your Pleiades are in the Bull's neck. The Hyades are the cluster near the eye.

What is the great star, a little out of the road, between Aldebaran and the Belt?

That is Betelgeux, in the shoulder of Orion, who is looking up to the Bull. Betelgeux is on the north side of the Belt; and Rigel, of Orion’s heel, is at the same distance to the south of the Belt. Aldebaran, Rigel, and Menkar, of the Whale’s nose, form an equilateral triangle.

Aries the Ram is to the right of the Bull, near where his tail should be, and Orion is to the left or east, but a little southerly. But now for the Twins.

Are they the two right north of Procyon, and forming the corner of a right-angled triangle with Betelgeux and Procyon?

They _are_ the Twins, Castor and Pollux, though the latter is the larger. Cancer the Crab is to the left of the Twins.

I cannot see so much as the claw of him. But I see a splendid star further to the left, with Procyon between it and the Belt.

That is Regulus in Leo the Lion. It is one of his forepaws, and beyond to the left is Deneb, of the Lion’s tail. The little lion is north of Leo, and between it and the Great Bear, the favourite constellation of England.

Virgo the Virgin should come next. I can see Spica, which is twice as far from the Cross northward as the Cross is from the Pole.

Spica is in the ear of corn held in the Virgin’s hand. The Virgin’s head touches the Lion’s tail at Deneb.

What a tall lady she must be! But what is that staring star forming the head of a triangle with Deneb and Spica?

The Arcturus of Job, in the knee of Bootes the Husbandman, standing on Mount Moenalus. The small stars between Arcturus and Deneb are in the hair of Berenice. The Northern Crown, between Hercules and Bootes, is seen nearly overhead in England.

You have got a long way from us now, father. Please come back to the Virgin.

Well, then, you know the Southern Crown, at any rate. Sagittarius is right over it northward, while Antares of Scorpio is to the right a little.

The stars of Libra the Balance must be somewhere between Spica and Antares, I suppose.

The two stars together in the equator, just north of Scorpio, are in the old man Ophiuchus, who is south of Hercules. There is a serpent on each side of him.

What nice company!

Yes, with the club of Hercules shaking over his head, and Poniatowski’s Bull ready to toss his shoulder. But he has the shield of the Polish hero Sobieski behind him, which hangs, too, over the head of Sagittarius.

How shall I find Aquarius?

He has his foot on Fomalhaut, he leans back on Capricornus, and has Pisces, the Fishes, on the other side.

What is the great star north of Capricornus?

That is Altair, in Aquila the Eagle, whose wings are spread over the equator. Aquila is between Ophiuchus and Aquarius.

As we have got over Aquarius, and come to Pisces, which are over Cetus the Whale, I suppose we are at Aries again.

We are. But I should like you to know that Pegasus, the Flying Horse, is over Aquarius, and that the Swan and Lyre are between that and Hercules, and north of the Eagle. Taking a line from the south pole through the Whale's tail, we have a bright star in the back of Pegasus, called Algenib, not far from the junction of the equator and ecliptic in Pisces.

The star beyond Algenib, further north, is larger, though.

Alpherat is its name, and it is one of the eyes of Andromeda, the lady with the chains on her hands.

O, I know all about that story, father. The ancients said she was fastened to a rock in the sea, and a monster was gnawing at her. But a warrior, called Perseus, came with his sword and delivered her.

Then, as you have Andromeda's eye, you can fancy the Zodiacal Fish biting her, and Perseus with his sword to the left. There is a triangle between Aries and Andromeda.

What a dazzling star that is, forming the north corner of a triangle with Aldebaran and the Twins.

Yes, Capella, or the Little Kid, in the arms of Auriga, is certainly very fine. Auriga's foot is on the Bull's head. Half-way from Alpherat of Andromeda to Capella, is Medusa's Head, just over the Pleiades.

We have gone round, I think, father, but I do wish I could see the old Bear again.

You have only to imagine yourself following out a line from the Belt through Castor and Pollux.

Yes, but I should have to dive down into the ocean to reach his claws."

DOUBLE STARS.

"That wonderful tell-tale, the 'telescope,' reveals the double character of some stars," said Mr. Marple. "There are six thousand stars which are thus double.

That is singular.

There are some in the Pleiades, in the Dog, in Orion, and in the Southern Cross and Crown. But the most wonderful part of my story is yet to come. Some of these double stars revolve round each other.

What! Great suns at a tremendous distance from one another, walk round one another? Why, if they have planets round them, like our sun has, these must have a fine double journey to make. How many of these strange stars are there?

About 650 of them are Binary or revolving.

But which stands still to have the other revolving round it?

They both move, each one round the other.

A regular polka in the heavens, father. And yet I suppose they are not just like a gentleman and his lady partner.

I don’t know that; for it has been observed that a number have different colours to their companions. While one is blue, the other will be yellow. One shall be green, and its mate will be orange. One-half are white, and one-tenth are both blue.

That does please me. How I should like to be in a planet where my sun should be blue, and its companion sun red. It would be so funny to get up in the morning and see my sun blue, throwing a blue tint on everything. Then, when the other sun rose, a rich red would steal over the blue, and change everything. But when the blue sun had set, the red one would have it all its own way, and bathe my world in its ruddy light.

That fancy is good, and not very improbable either; especially if the other sun should come pretty near your sun at times, as it is likely they move in an ellipse.

Can you tell me, dear father, how long these pleasant binary stars take in performing their polka?

One in Hercules takes thirty-one years. In our old Great Bear, a binary pair take sixty-one, and in Leo eighty-two. In the Swan we have 178 years, and in the Virgin 182. In the Crown we have two several times given of 608 and 736 years. It is believed that one pair are 1200 years. The North Polar Star is double, and is thought by some to revolve in 6000 years.

Please, father, tell me some more about the colours of the double folk.

The Polar double stars are red and white. In Scorpion, there is a white one and a lilac one; in the Centaur, we have both orange; in Andromeda, crimson and green; in the Lion, gold and green; in one of Scorpio, White and plum; in Castor, yellow and green; in Antares, both are bright red; Arcturus was all red in 1841, and yellow in 1852.

I wonder how far these partners are from each other.

That is difficult to ascertain. The binary stars of the Swan are thought to take 450 years going their circuit, and yet to be billions of miles away from each other.

There is little danger of quarrelling at that distance apart, father.

Astronomers talk of stars running against one another. There are also triple stars.

Three stars polkaing in and around each other! That is odd. Where are there any?

In the Lyre and Scorpion. In the latter the colours are white, grey, and yellow. In the Swan the largest star is very yellow, and its two mates are blue. But more wonderful still, in Orion there are four stars thus rolling round each other."

PERIODICAL AND VARIABLE STARS.

It was upon a lovely evening, when off the coast of South America, ready to slip across by the side-help of the South-east Trades to the Cape of Good Hope, that Mr. Marple found his son at his old post.

"Now then," said he, "my star-gazing boy, are you quite sure that none of them have run away since last night?

I am not sure of that, father; for ever since you proved that the Fixed stars were not fixed, I think I might some day find a bright friend gone on a visit to another universe somewhere; perhaps, getting into such agreeable society as to have no wish to come back again to our visible universe; or, maybe, to pop back again in his place.

It so happens, James, that what you have imagined, really takes place. Some stars do really disappear, to return, or not; and others, wholly strangers to us, come within observation.

That is the queerest thing you have told me yet. What droll fellows the stars are, after all! How they do wriggle about, polkaing here, doubling there, and fielding about like boys at cricket. I shall never be surprised at any funny trick you ascribe to them.

To begin with the _Variable stars_, of which there are known about one hundred. There is Mira, which is invisible five months in the year. It goes and comes again.

Where could it have got to?

Some think it revolving round a dark object; and, of course, becoming invisible when on the other side from us. One passes through three magnitudes in one month.

But I must give you the history of such a curious star in the Whale. For two weeks you might see it a noble star of the second magnitude. It gradually dims, till it goes fairly out of sight in three months. For five weeks you see nothing of it. It then reappears. Altogether it seems to have a revolution of 322 days; and is, therefore, one of the _Periodical_ stars.

I should call it a variable periodical.

There is another in Cepheus, which grows to its full light in thirty-eight hours only. There is one in the Lyre taking ninety-one hours. Some fancy that the dark appearance may be owing to huge spots on the suns coming into view as the suns turn round. But there is still another sort of star--the _Temporary_.

Do they come and have a look at us in their birth, and then disappear never to return?

I will tell you the history of one. In 1572 an astronomer saw it shining in the daytime. He could not find it in his star catalogue at home. Day after day it grew

brighter and brighter, till it blazed far more than either Sirius or Jupiter, and could be plainly seen in the noontide sun. It then grew dim gradually, and in two years went away altogether, without returning since.

That is temporary, and no mistake. But still I guess they will find him turning up some day and becoming a regular Periodical.

There is a wonderful star, Argus, in Argo. Two hundred years ago it was of the fourth magnitude only. In 1834, Sir John Herschel, when at the Cape of Good Hope, found it a second magnitude. The light was three times stronger at the end of 1837. Next month it was brighter than any star here but Sirius. It faded awhile, and then increased, till, in 1844, it was equal to Sirius itself.

Have they looked over the old catalogues of stars, made, as you told me, two thousand years ago, just to see what ones have dropped out of sight, father?

Yes, and several have disappeared which were on that list."

MILKY WAY.

The stars were out in their very best humour, looking so clear and merry in their twinkling; while the Milky Way stole out to view in all its maiden, lily beauty. Can we wonder, then, that our young voyager sang out--

"O what a splendid starlight night! There seems to be ever so many more stars than usual. Can any one count them all?

Yes; those you now see are about 2000. But the telescope shows us far more. There are 50,000 regularly catalogued. A magnifying power of 180 would give 20,000,000. A small patch of the Milky Way has revealed millions of stars.

Milky Way, father! O that is that serpent-like twisting of white stuff across the heavens. But why should it look so white and all other parts so black?

Because it is there that most of the stars lie, and give out the brightest light.

But I don't see that the milk is all stars.

Take up the telescope and look at that white patch near that red star.

What a pretty sight! Why, a lot of the milk has turned into stars already. I am sure there must be a tremendous lot to be seen through the telescope of an astronomer.

Do you remember the Milky Way on the northern side of the equator?

Yes, father, and I think it a prettier looking white snake in the sky than this one up here, for it had more spangling stars about it.

I must say that the northern constellations are more brilliant than those of the south.

And what a great bit of the south side has hardly any stars at all. Just look, father, there to the westward, and there to the eastward. They both look as if lighted up with a few children's toy-candles, compared to the dazzling light of the north, and right overhead toward the south.

The Milky Way is narrowest near the Southern Cross, and widens both ways after. Near the Scorpion’s tail it is very broad--about twenty degrees, while at the Cross it is about four degrees. There is a curious opening in it near Sirius, like a long, narrow, dark lake among the mountains of snow.

And I can see one star at the end, like a ship in full sail upon the lake.

The Milky Way runs up in threads near the Scorpion. From the Swan down to the Southern Cross it is in two bright lines, enclosing a long dark space.

But look at the two pretty bridges over the black waters, connecting the two shining walks at the sides.

You remember, then, that the Milky Way, seen in the north and south, is a great ring of different breadths going round the heavens, and passing at some little distance from both poles.

It is like the Egyptian story you told me, father, of the people worshipping a serpent with his tail in his mouth. How singular that this milk-stream should be poured right round the heavens in a huge broad circle!

The Galaxy, or Milky Way, is seen to great advantage near the Cross.

It is about the milkiest in that quarter. And I can make a good guess now about the two Magellanic clouds, as the captain called them--these two clouds of light not far from the Cross. Why, they are only places where the shining stuff is thickest.

You would, then, really believe there might be lots of stars in those clouds.

Yes; but what are the two black clouds?

Look at them attentively, and tell me what you think of them?

There is no milk there, anyhow. I suppose they look black by contrast with the bright clouds near. How cold and dreary they seem! And yet now I can make out a few stars scattered about, like ships on a big sea at a great distance from one another.

The early navigators southward used to call them the _Coal Sacks_, because so black.

Why, the Southern Cross seems to rise from that pear-shaped Coal Sack, and all at once to get crowned with a blaze of light.

True, my dear boy. The Cross of our faith has often brought man from the darkness of doubt and misery into the glorious light of freedom and hope.

Thank you, kind father, for saying that. I do trust I may always have the cross shining on my way through life.”

THE NEBULÆ.

How wondrous a sight did James think the Milky Way! Evening after evening would he stare at the weird-looking object. Moonlight eclipsed it, as sunlight the stars. The atmosphere through which we contemplate the heavens at times obscures their glory.

That dreamy radiance is easily concealed from view. The bold planet looks down upon us with an unblinking eye. The fixed stars peep more coyly, with an uncertain lustre. The Milky Way, yet more retiring, seldom deigns to do more than glance timidly at us here. With our magnificent climate in Australia, we are favoured beyond most countries with its soft, peculiar light.

“What is a Nebula, father? said James.

A white, cloudy patch in the heavens. There are about five thousands of them scattered about space.

You said the Milky Way was made up of stars and Nebulæ, and that many patches of white, when looked at with a good telescope, turn out to be only lots of stars.

Yes. These are the _Resolvable_ Nebulæ. But there are _Irresolvable_,--that is, some of them still look cloudy with our best telescopes. Can you find me a Nebula in Orion?

What must I look for?

A small, distinct, white patch of light.

No; I don’t see it.

Look again. Do you see the Belt, with a bright star over it, and one about the same distance below it?

I see all that clear enough, but not your Nebula.

Look, boy, steadily and closely between the Belt, and that top bright star Rigel, where on the Atlas is pictured the sword-handle of the great hunter.

I don’t feel very sure, but I fancy I can distinguish something that ought to be it.

Many folks, my boy, can see what they are told to expect. Well, that white irregular patch of light has had telescopes looking at it a good while, without anything being seen beyond a dreamy-looking cloudy matter.

I’ll be bound Lord Rosse solved the riddle.

He did not for some time. He was able at last to think he could see stars; then by more patient watching he resolved the Nebula into sandheaps of stars--millions upon millions.

What! millions upon millions where other telescopes could not distinguish one star. What shape is Orion’s Nebula?

It is rather patchy, with innumerable streamers of light, as if wind were blowing the gauze stuff about in all directions. You might fancy in it the jaws and head of a monster, with an elephant’s proboscis.

What a nice little nose that must be.

One part rises like a conical cloud in the midst of the black sky. In the part which had appeared mottled Rosse found a blaze of stars.

Can all the Nebulæ be observed by the naked eye?

No, my lad, very few.

But after all they are only lots of stars got crowded together like, because they are so far from us.

Yet there are Nebulæ not to be resolved into stars even by Rosse's six feet mirror. Just turn round to the Southern Cross. You see the two bright Pointers to the Cross, a part of the Centaur. Look now to the other side of the Cross, where there is a collection of stars scattered about. That is the _Argo_ or ship.

Ah! I can see a sort of light there. Is that the Argo Nebula?

Yes, and a very large one it is. Thousands of stars can be observed in it by the telescope; but beyond these is still the same filmy light, as irresolvable as ever.

There must be a lot of Nebulæ by the Cross, judging by the blaze of light.

There is a very fine one there, of a blue colour. One near Spica, of the Virgin, is quite round, and of enormous size. But there is a very odd Crab Nebula between Orion and the Bull's eye. It branches out like the claws of a crab.

What an enormous Crab!

We have a Dumb-bell Nebula near the Lyre.

Oh! who could swing that about? Pray tell me some more about these queer creatures, father.

Many in the south are planetary Nebulæ, as they are in the midst of a cluster of stars. They are always of a pale blue colour. Some are double ones. About a dozen Nebulæ are annular, or ring-shaped. Some are double or treble ringed. Others have rings within rings, and star-like eyes in the darkness. A few have a long, parsnip form. There are solid Nebulæ, and hollow ones. One near Castor was found by Rosse to be arched like eyebrows above and below, with a curious, crab-like form in the middle.

I wonder what it would look like should a better telescope reach it!

Wonder, indeed! Some Nebulæ seen by Herschel get quite another shape with Rosse. The six-feet mirror gives another look to that by the three-feet one.

Then I can scarcely believe the present shapes, as a better instrument may resolve the thing into another creature altogether. But is there any other sort beside the globular and annular?

Yes; a good many are spiral, or corkscrew form.

Something like a shaving that a plane fetches off a plank, father.

But some of your celestial shavings have a wild look. Clever men have fancied that the star-dust is rolling up that way by a sort of gravitation into a regular form.

What of the two clouds in the heavens opposite to the Southern Cross, down in the south? They look uncommonly like monster Nebulæ.

And monsters indeed they are. In the larger one of them three hundred Nebulæ and globular clusters of stars have been distinguished, and fifty in the smaller one.

Then there is room enough in the Cloud for a whole universe?

Yes, when we find the larger Magellanic Cloud, or Nubecula Major, as it is called, taking up many times the space occupied by the sun in the sky. The Nubecula Minor, or Little cloud, is about one-fourth the size of the larger.

What do you mean by Nubecula?

A little cloud. They are called the Magellanic clouds because first noticed in the voyages of Magellan the Portuguese, three hundred years ago.

What a space they must take up in reality!

Especially if, as astronomers believe, the clusters of stars and patches of nebulous matter in the two clouds there are as far off from each other as from this earth.

What is known about the Coal-Sacks?

The larger Coal-Sack, or black empty space not far from the two clouds, is of a pear shape, and occupies a space of eight degrees long by five broad. Its darkness is only comparative, as two hundred stars have been noticed therein by a good telescope.

Are the Nebulæ of the Clouds peculiar?

One in the centre is very large, and has an odd dark space in its middle. It is also surrounded by a circle of ten other Nebulæ.

Like bright guards around a king.

Another consists of four starry centres, or nuclei, which are curiously united by a very faint nebulous matter.

Do, father, tell me more of these Nebulæ.

Then I will talk about some singular filaments, or threads of light, seen around the nebulæ, which have much puzzled astronomers. Those filaments about the Nebulæ of Orion and Argo are wonderfully strange.

They are the arms, I suppose, pointing into the dark space near, as much as to say,--‘You know nothing about us, or of what we can see around us, looking so black to you, and so bright to us.’

Not a bad idea, boy. When Sir John Herschel had a look at our southern hemisphere, he observed beyond the Nebulæ a few scattered, faint bits of light in the dark region, too fine for him to turn into Nebulæ by his glass, and looking like some more distant light blushing through the darkness.

They are like telegraphic messages, father.

About forty of these strange glimmerings of future universes yet to be revealed were seen by Sir John.

Well, dear father, you can’t go further than that. If it take 60,000 years for some of the stars to send us their light, just to say ‘How do you do?’ how long would it take for these faint, modest blushes of the distant sky?

How great, then, is God, my son, and how grateful should we be for His notice! Let us so please Him here, that after death our souls may fly to that heaven of joy He has prepared, wherever in the vast universes of His it may be.”

THE SUN AND PLANETS.

The planet Venus had been shining brilliantly evening after evening. “Why, father,” said the boy, “does Venus shift her place so rapidly among the stars?

Because she has to travel round the sun, as the world and other planets do.

And is the world really a planet, looking like a star to the people of Venus?

Yes; and not visible at all to those of the distant planets, which are all so much nearer to us than the nearest of the fixed stars. Some of the planets you could not see without the telescope, and you see none of the asteroids.

What are they?

Little planets, over eighty in number, perhaps the shattered parts of an exploded planet, travelling along the old road, and keeping a circle between Mars and Jupiter.

Then they are not like the comets, which run everywhere, getting in everybody’s way?

You are slandering the comets. These shadowy bodies, with tails of light, travel in longer ovals or ellipses than the planets. We see them when they turn round the sun, which is near one end of their orbit, but they disappear afterwards in the great distance of their curve.

Well, I _can_ see some use in the sun to us, but very little in the planets and other stars, and none in the comets.

You must not selfishly judge of the good of things by the use you can make of them. The extensive granite ranges were not made merely for man’s building materials, nor the stars for a gaze through a telescope. As the world existed many millions of years without man, so have the stars. In another world we may know more of the why and because of God’s creation.

May not the planets and other stars be inhabited by folks like ourselves?

Not quite, my son, replied Mr. Marple, though God can make intelligent beings to praise Him, who would be adapted to any atmosphere or condition of the stars.”

In the evening the captain lent James a book, from which he copied the following table of the Eight Planets--their distance from the sun, their times of revolution round the sun, and their size:--

Distance. Time. Diameter. Miles. Miles.

Mercury 37,000,000 2 months 3100 Venus 69,000,000 7-1/2 ” 7900 Earth 93,000,000 1 year 7920 Mars 146,000,000 2 years 4000 Jupiter 494,000,000 12 ” 92,000 Saturn 906,000,000 30 ” 75,000 Herschel 1820,000,000 80 ” 36,000 Neptune 2850,000,000 164 ” 33,600

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